

Charged-Polymer FTS: Compressible Smeared-Charge Theory

2026-08-02

Working notes (devel). Formulation of polyelectrolyte solutions within this codebase’s multi-monomer field theory, following the smeared-charge model of Wang [1] and Villet–Delaney–Fredrickson [2,3], written in the conventions of *Macromolecules* **2025**, 58, 816 (the neutral multi-monomer theory this library implements). **Compressible (finite ζN) model with per-species density smearing** — the fully UV-regularized formulation.

1. Model

Species $i = 1, \dots, S$ (here: polymer P, solvent S, counter-ion C, salt ions SP/SM). Each species is a (possibly very short) bead-spring chain of the discrete or continuous model already implemented.

Smeared densities. Every species carries a normalized Gaussian shape function

$$h_i(\mathbf{r}) = \frac{1}{(2\pi a_i^2)^{3/2}} e^{-r^2/2a_i^2}, \quad \hat{h}_i(\mathbf{k}) = e^{-a_i^2 k^2/2},$$

of radius a_i (params **radiuses**, in R_0 units). a_i is the **smearing length** (Fredrickson school: width of the “shape function” or form factor of the species [2,3]); for charged species Wang [1] identifies it with the ion’s **Born radius**, since the Gaussian self-energy $u_{\text{self}} = z_i^2 l_B / (2\sqrt{\pi} a_i)$ is the Born solvation energy of an ion of that size. ALL interactions are written in terms of the smeared volume fractions

$$\bar{\phi}_i \equiv h_i * \phi_i.$$

Smearing regularizes both the contact (excluded-volume/ χ) and the Coulomb self-interactions, making the fluctuating field theory UV-convergent [1,2,3]. The point-particle limit $a_i \rightarrow 0$ is singular for a compressible fluctuating model and must not be taken. ($\$a_i = \$ \text{None} \Rightarrow h_i = \delta$; acceptable only for testing or in mean-field.)

Three interaction channels:

1. **Flory–Huggins:** χN_{ij} pairs (here only P,S — all pairs involving ions are zero). No separate **chi_monomers** machinery is needed in the compressible model: the χN matrix is simply $S \times S$ with zero rows/columns for the ions (see §3).
2. **Compressibility (Helfand):** instead of the incompressibility constraint, a finite penalty

$$\beta U_\zeta = \frac{\rho_0 \zeta N}{2N} \int \left(\sum_i \bar{\phi}_i(\mathbf{r}) - 1 \right)^2 d\mathbf{r}$$

with the code’s existing **zeta_n** parameter. The incompressible theory is the $\zeta N \rightarrow \infty$ limit. In a fluctuating simulation the smeared $\bar{\phi}_i$ (not raw ϕ_i) must enter here, or the contact penalty is UV-divergent.

3. **Coulomb**: segment of species i carries per-segment valence z_i ; the smeared charge density is

$$c(\mathbf{r}) = \sum_i z_i \bar{\phi}_i(\mathbf{r}) \quad (\text{same } h_i \text{ as above}),$$

and

$$\beta U_C = \frac{l_B \rho_0^2}{2} \iint \frac{c(\mathbf{r}) c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}', \quad l_B = \frac{e^2}{4\pi\epsilon k_B T}$$

(Bjerrum length; uniform dielectric ϵ , no dielectric contrast in this first version).

Global electroneutrality is still required (compressibility does not fix charge). With per-segment valences z_i and equal segment volumes:

$$\sum_i z_i \bar{\phi}_i^{\text{tot}} = 0$$

over overall volume fractions (NO division by chain length — that would be per-chain counting, wrong for per-segment valences). The counter-ion fraction is NOT a free parameter: for P with $z_P = +1$ and fraction ϕ_P , one needs $\phi_C = \phi_P z_P / |z_C|$ (plus balanced salt $z_{SP}\phi_{SP} + z_{SM}\phi_{SM} = 0$). *The current **Lamella.py** placeholder sets $\phi_C = 0$ with charged P — not electroneutral; the $k = 0$ Coulomb mode diverges. The driver must compute ϕ_C from ϕ_P .*

2. Units and code conventions

Same as the neutral theory: lengths in $R_0 = a_{\text{Ref}} N_{\text{Ref}}^{1/2}$, fields per **reference chain** ($w_{\text{code}} = N w_{\text{per segment}}$), densities in ρ_0 . Chain number density $C = \rho_0 R_0^3 / N = \sqrt{N}$.

Nondimensionalize with $\mathbf{x} = \mathbf{r} / R_0$. The Coulomb energy in $k_B T$ becomes (note the C^2 — Coulomb is quadratic in ρ_0 , unlike the χ/ζ block where $\chi N, \zeta N \propto \rho_0$ leave one C):

$$\beta U_C = \frac{C^2 E_0}{8\pi} \iint \frac{c(\mathbf{x}) c(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d\mathbf{x} d\mathbf{x}' = \frac{C^2 E_0}{2} \sum_{\mathbf{k}} \frac{|\hat{c}_{\mathbf{k}}|^2}{k^2}, \quad E_0 \equiv \frac{4\pi l_B N^2}{R_0}.$$

For the per-chain-normalized Hamiltonian used throughout this codebase, the convenient coupling is

$$E \equiv C E_0 = \frac{4\pi l_B N^2 C}{R_0} = 4\pi l_B \rho_0 N R_0^2$$

(density-DEPENDENT: $E = \sqrt{N} E_0$). Caveat: the literature's “electrostatic strength” [2,3] is the density-independent E_0 , usually in $R_g = R_0 / \sqrt{6}$ units — convert when comparing. New scalar param: “bjerrum_e” = E (or “bjerrum_length” = l_B / R_0 with E computed internally from `nbar`).

3. Field theory: Hubbard–Stratonovich structure

Two independent HS blocks (the compressible model has NO separate pressure constraint):

(a) $\chi + \zeta$ block — the existing compressible SPT, over **ALL** S species. The quadratic form of channels 1+2 is

$$\frac{1}{2} \bar{\phi}^T (\zeta N \mathbf{J} + \chi N) \bar{\phi} - \zeta N \mathbf{1}^T \bar{\phi} + \frac{1}{2} \zeta N,$$

with $\mathbf{J}$ the all-ones matrix — exactly the matrix $u = \zeta N \mathbf{J} + \chi N$ that `SymmetricPolymerTheory`'s compressible branch (`zeta_n` set) already diagonalizes, including the linear term (its `vector_large_s`).

Ion rows of χN are zero. Spectrum for the 5-species example: - one large positive eigenvalue $\approx S\zeta N$ (the “pressure-like” mode) — imaginary-type; - the χ -split P–S pair: one negative (exchange, **real** Langevin field), one positive (imaginary-type); - pure ion-composition modes orthogonal to **1** with no χ : eigenvalue **0** — ideal mixing in those channels; the corresponding auxiliary fields are identically zero and SPT’s existing zero-eigenvalue handling (Gram–Schmidt + warning) applies. This is physical, not an error: those composition fluctuations are free.

Because $\zeta N \mathbf{J}$ lifts the would-be singularity of the χ -only matrix, **the full S -species SPT can be used directly** — the chi-subset wrapper of the incompressible draft is no longer needed.

(b) Coulomb block — new. HS on the positive-definite Coulomb quadratic form introduces the electrostatic potential $\psi(\mathbf{x})$:

$$e^{-\beta U_C} = \int \mathcal{D}\psi \exp \left[-C \int d\mathbf{x} \frac{|\nabla\psi|^2}{2E} + i C \int d\mathbf{x} c(\mathbf{x}) \psi(\mathbf{x}) \right] / \mathcal{N},$$

which reproduces $\exp[-\frac{C^2 E_0}{2} \sum_k |\hat{c}_k|^2 / k^2]$ exactly because $E = C E_0$ (Gaussian integral over ψ). ψ is the per-reference-chain potential $N\beta e\varphi_{\text{phys}}$. Coupling with $+i$ to a real density and a positive-definite kernel ($4\pi l_B/k^2$, analogous to $\lambda > 0$): ψ is an **imaginary-type field**, partial-saddled like the positive-eigenvalue SPT fields — see §5.

4. Hamiltonian and single-chain problem

The per-chain effective Hamiltonian is

$$\frac{H[\{w_k\}, \psi]}{C k_B T V / R_0^3} = \underbrace{h_{\text{const}} + \sum_k [A_k \langle w_k \rangle + B_k \langle w_k^2 \rangle]}_{\text{neutral compressible (existing SPT coefficients)}} + \frac{1}{V} \int \frac{|\nabla\psi|^2}{2E} d\mathbf{x} - \sum_p \frac{\bar{\phi}_p}{\alpha_p} \ln Q_p[\{W_i\}]$$

with the h -coefficients of the existing compressible SPT. The one-body potential of each species picks up the smearing convolution on EVERY term (the interactions couple to $\bar{\phi}_i = h_i * \phi_i$, so the functional derivative w.r.t. ϕ_i convolves back with h_i):

$$W_i(\mathbf{x}) = h_i * \left[W_i^{\text{SPT}} + i z_i \psi \right](\mathbf{x})$$

where W_i^{SPT} is the usual **matrix_a** mapping of the auxiliary fields (now including the pressure-like ζ mode; ion rows couple only to that mode and to ψ). No explicit N multiplies the electrostatic term: ψ is already per-reference-chain and z_i is per-segment — the code’s $w = N w_{\text{seg}}$ convention is carried entirely by ψ . The smearing is a k-space multiplication $\hat{h}_i(k) [\widehat{W_i^{\text{SPT}}} + i z_i \hat{\psi}](k)$ — cheap, per species, applied once per saddle iteration; the propagator solvers are untouched.

Rotated contour / stored real field. As with the positive-eigenvalue SPT modes, the code stores the REAL array $\psi_s = -i\psi$ (the saddle lies on the imaginary ψ axis). Under this substitution the one-body term becomes $+z_i \psi_s$ (the i disappears — this is what the code adds to W_i^{SPT}), and the explicit quadratic term flips sign:

$$\frac{1}{V} \int \frac{|\nabla\psi|^2}{2E} d\mathbf{x} \rightarrow -\frac{1}{V} \int \frac{|\nabla\psi_s|^2}{2E} d\mathbf{x} = -\frac{1}{2V} \int c \psi_s d\mathbf{x}$$

(the last equality at the Poisson-consistent ψ_s). This mirrors the negated B_k coefficients of the imaginary SPT fields. Note the $+c\psi_s$ coupling itself lives inside $-\ln Q$ via W_i ; the explicit term recorded in H is only the (negative) gradient piece.

Concentrations: standard ϕ_i from propagators; the smeared densities that enter all interaction terms and the Poisson source are $\bar{\phi}_i = h_i * \phi_i$, $c = \sum_i z_i \bar{\phi}_i$ (again diagonal in $\mathbf{k}$).

5. Forces and saddle conditions

- SPT auxiliary fields: unchanged from the neutral compressible theory — functional derivatives evaluated with the SMEARED densities $\bar{\phi}_i$ in place of ϕ_i . Negative-eigenvalue (exchange) fields receive Langevin noise; positive-eigenvalue fields are partial-saddled by the existing compressor loop; zero-eigenvalue fields stay zero.
- Electrostatic field:

$$\frac{1}{C} \frac{\delta H}{\delta \psi(\mathbf{x})} = -\frac{\nabla^2 \psi}{E} - c(\mathbf{x}) \quad \Rightarrow \quad \text{saddle: } -\nabla^2 \psi^* = E c(\mathbf{x}),$$

the (smeared) **Poisson equation**; with the ideal-gas ions responding through their Boltzmann factors this is the fluctuating-field generalization of Poisson–Boltzmann. In $\mathbf{k}$ -space the solve is diagonal and exact:

$$\hat{\psi}^*(\mathbf{k}) = \frac{E \hat{c}(\mathbf{k})}{k^2}, \quad k \neq 0,$$

and the $k = 0$ mode is fixed by electroneutrality ($\hat{c}(0) = 0$ holds exactly per configuration in the canonical ensemble once §1’s constraint holds; set $\hat{\psi}(0) = 0$ as gauge).

Debye sanity check (1-segment ions, $z = \pm 1$, ideal): linearizing gives $\kappa^2 R_0^2 = E \bar{\phi}_{\text{ion}}/N = 4\pi l_B \rho_0 \bar{\phi}_{\text{ion}} R_0^2$ — the physical Debye constant. Any factor error in E or W_i shows up here first; keep this as a unit test.

CL-FTS (the real implementation): ψ fluctuates. L-FTS holds imaginary-type fields at partial saddle, so it can never capture ψ fluctuations — the charged theory belongs in CL-FTS, where ψ evolves with complex Langevin dynamics exactly like the pressure field W_+ (rotated storage, imaginary-direction noise, “ $+\Lambda$ ” drift):

$$\psi \leftarrow \psi + \left(\frac{\nabla^2 \psi}{E} + c \right) \Delta t s_\psi + i \mathcal{N}(0, \sigma) \sqrt{s_\psi},$$

whose drift fixed point is the Poisson equation. Three discrete-scheme facts (all found the hard way):
- The ∇^2/E part is stiff at high k : explicit Euler is UNSTABLE ($k^2 \Delta t/E$ exceeds the stability limit), and semi-implicit Euler, while stable, SUPPRESSES the stationary variance by $1/(1 + 2\kappa \Delta t)$ per mode — this silently wrecks fluctuation thermodynamics (the one-loop $\langle \partial H / \partial E \rangle$ came out 2–4 $\times$ too small). The correct treatment is **ETD (exact Ornstein–Uhlenbeck integration per $\mathbf{k}$ -mode)**:

$$\hat{\psi}' = a_k \hat{\psi} + (1 - a_k) \frac{E \hat{c}}{k^2} + i \sigma \sqrt{\frac{1 - a_k^2}{2\gamma_k \Delta t}} \hat{\xi}, \quad a_k = e^{-\gamma_k \Delta t s_\psi}, \quad \gamma_k = k^2/E,$$

which reproduces the exact per-mode stationary variance of the linear part for ANY Δt ; the $c[\psi]$ screening stays explicit (per-step rate $\leq \sum_i z_i^2 \bar{\phi}_i \Delta t$). The $k = 0$ mode is gauged to zero every step. - The EXCHANGE field’s explicit mass is $\sim 1/\chi N$, so small- χ systems (e.g. the pure-salt validation) need $\Delta t < \chi N$; violating this diverges regardless of the charge sector. - The explicit Hamiltonian term must be evaluated ANALYTICALLY off the saddle: $+(1/2EV) \int \psi \nabla^2 \psi$ (equal to $-\frac{1}{2} \langle c\psi \rangle$ only when Poisson holds).

At $E = 0$ the ψ stiffness is infinite: ψ stays pinned at zero.

L-FTS prototype (mean-field ψ): scheme per saddle iteration — after each propagator solve, (i) build $\bar{\phi}_i$, c (k-space $\hat{h}_i$ multiplications), (ii) update ψ toward the Poisson solution (below), (iii) take the usual Anderson-mixing step on the imaginary SPT fields. Both residuals enter the stopping criterion. ψ stays OUT of `aux_fields_imag_idx` and the AM compressor state.

Stabilized ψ update (important). Replacing ψ by the bare solution $E\hat{c}/k^2$ every iteration is an undamped fixed-point sweep whose linearized gain is

$$g(k) = \frac{E}{k^2} \sum_i z_i^2 \bar{\phi}_i \hat{h}_i^2(k),$$

which exceeds 1 at long wavelengths for realistic E (e.g. $E = 25$, $k_{\min} = 2\pi/L$) — the joint saddle loop then diverges. Use instead the preconditioned Newton update with the ideal (local) screening estimate $S_{\text{scr}}(k) = \sum_i z_i^2 \bar{\phi}_i \hat{h}_i^2(k)$:

$$\hat{\psi} \leftarrow \hat{\psi} + \frac{E\hat{c} - k^2\hat{\psi}}{k^2 + E S_{\text{scr}}(k)}, \quad \hat{\psi}(0) = 0.$$

The fixed point is unchanged ($-\nabla^2\psi = Ec$, monitored via the residual $-\nabla^2\psi/E - c$), but the screening term in the denominator damps exactly the modes the bare sweep amplifies; for ideal 1-segment ions the update is a true Newton step. S_{scr} only sets the convergence rate, so the segment-fraction estimate suffices for polymeric charges.

6. Discretization notes

- All new operations are diagonal in k-space: $\hat{h}_i(k)$ multiplications and the E/k^2 Poisson solve. Use the existing FFT objects and the deformation-vector $|\mathbf{k}|^2$ tables (**Pseudo**) — oblique cells work automatically.
- **No real-space $\nabla\psi$ is ever needed.** The electrostatic energy is evaluated by Parseval, and at the saddle the substitution $\hat{\psi} = E\hat{c}/k^2$ eliminates the gradient entirely:

$$\frac{1}{2E} \int |\nabla\psi|^2 d\mathbf{x} = \frac{E}{2} \frac{V}{M^2} \sum_{\mathbf{k} \neq 0} w_t \frac{|\hat{c}_{\mathbf{k}}|^2}{k^2} = \frac{1}{2} \int c \psi d\mathbf{x}$$

(unnormalized `rfft`; $w_t = 2$ interior / 1 edge modes, as in `wtmd.py`). Only $h_i * \psi$ requires an inverse transform. A real-space gradient (component-wise $ik_\alpha \hat{\psi}$) first becomes necessary for the deferred Maxwell-stress term. The identity above gives the MAGNITUDE; on the rotated contour the term entering the recorded H is $-\frac{1}{2V} \int c \psi_s$ (see §4) — the code evaluates it as `-0.5*mean(c*psi)`.

- Smearing widths must satisfy $a_i \gtrsim dx/2$ **per species** — the binding constraint is $\min_i a_i$; below that the regularization becomes grid-dependent (same issue class as the bond-function study). With finite ζN this applies to ALL species, not just charged ones.
- Per-species radii compose automatically: the effective i - j pair kernel is $\hat{h}_i(k)\hat{h}_j(k) = e^{-(a_i^2 + a_j^2)k^2/2}$, i.e. an effective pair smearing width $\sqrt{a_i^2 + a_j^2}$ — Gaussian widths add in quadrature, so mixed radii (small ions, fat polymer segments) need no special treatment. Consequence for validation: with unequal radii the RPA/Debye–Hückel comparison must keep the per-species $\hat{h}_i^2$ factors separate (they only factor out as a common $\hat{h}^2$ when all radii are equal); the $k \rightarrow 0$ Debye limit is radius-independent ($\hat{h}_i(0) = 1$), which is why it is the safe unit test.

- Charge-charge structure factor $S_{cc}(k)$ from $\hat{c}(k)$: record alongside the existing $S(k)$.
- Stress with electrostatics (Maxwell term from $\partial_L \int |\nabla\psi|^2/2E$) and with the $\hat{h}_i(k)$ cell dependence: defer; raise on `box_is_altering=True`.

7. Parameter dictionary (target)

```
"zeta_n": 100.0,                                # compressible Helfand penalty (existing param)
"charges": {"P": 1.0, "S": None, "C": -1.0, "SP": +1, "SM": -1.0},
"radiuses": {"P": 0.025, "S": 0.025, "C": 0.025, "SP": 0.025, "SM": 0.025},
"bjerrum_e": 10000.0,                           # E = 4 pi l_B rho_0 N R O^2 (NEW)
"chi_n": {"P,S": 50},                           # full-matrix SPT; ion rows zero
# NO "chi_monomers" needed; NO incompressibility constraint
# molecules: counter-ion fraction computed from electroneutrality, not free
```

None charge = neutral species (no ψ coupling). **radiuses** now apply to ALL interactions (χ , ζ , Coulomb) of that species — in the compressible fluctuating model neutral species need smearing too.

8. Implementation (devel-only; CL-FTS is the primary implementation)

Where the code lives (2026-08-02): entirely in THIS folder — mainline `src/` is untouched. - `electrostatics.py`: per-species smearing kernels (real AND complex fields), coupling E , electrostatic Hamiltonian, electroneutrality check, plus the ψ Newton update used by the L-FTS prototype. - `clfts_charged.py` — **the primary implementation**: `ChargedCLFTS(polymerfts.clfts.CLFTS)` with ψ as a fully fluctuating CL field (§5): overrides `compute_concentrations` (complex $W_i = h_i * (W_i^{\text{SPT}} + z_i\psi)$) and `run` (co-evolves ψ semi-implicitly, adds $-\frac{1}{2}\langle c\psi \rangle$ to H , saves ψ in checkpoints). ψ noise uses a separate PCG64 stream (seed+1) so the neutral reduction stays bit-exact. - `lfts_charged.py` — mean-field- ψ PROTOTYPE kept for validation only: its deterministic saddle makes the sampling-free Debye-Hückel linear-response test possible (`test_debye_huckel.py`), which pins the E normalization/kernels shared with the CL class.

L-FTS samples only the REAL exchange fields and holds imaginary-type fields at partial saddle, so ψ there is mean-field (Poisson-Boltzmann level) — that is why CL-FTS is the real home of the theory. The compressible choice simplifies everything versus the incompressible draft:

1. `polymer_field_theory.py`: **unchanged** — the compressible branch (`zeta_n`) over the full S -species matrix already produces the right eigen-system, including the zero-eigenvalue ion modes (existing handling).
2. Relation to the existing global `Smearing` class: that class implements the SPECIAL CASE of one common radius applied to all fields — the prototype bypasses it (adapter), never applying both to the same field.
3. Ions as short chains work today (`length=0.01` $\Rightarrow$ 1 discrete segment); no C++ changes — everything new is k-space Python.
4. Validation ladder:
 - a. $E \rightarrow 0$, uniform a_i : must reproduce the existing neutral compressible run with global smearing exactly.
 - b. $\zeta N \rightarrow \infty$, $E \rightarrow 0$: approach the incompressible neutral results.
 - c. Salt-only (no polymer): $S_{cc}(k)$ against smeared Debye-Hückel/RPA, $S_{cc}^{-1} = [S_{cc}^{\text{ideal}}]^{-1} + E/k^2$ where S_{cc}^{ideal} is the ideal structure factor of the SMEARED charge density (contains

the $\hat{h}_i^2$ factors — with a bare ideal S one must write $\hat{h}^2 S / (1 + (E/k^2)\hat{h}^2 S)$; the $\hat{h}^2$ cannot be dropped). $k \rightarrow 0$ recovers the Debye κ^2 of §5.

- d. Literature anchor: polyelectrolyte solution structure of Refs. [2,3] (mind the E_0 vs E and R_g vs R_0 conversions of §2).

Validation status (2026-08-02): - (a) Neutral reduction PASSED bit-exactly for BOTH classes (`test_neutral_reduction.py` for `ChargedLFTS`, `test_neutral_reduction_clfts.py` for `ChargedCLFTS`): $z = 0$ with uniform a_i reproduces the neutral run with global smearing to the last bit over 10 Langevin steps. - (c) Debye–Hückel PASSED. Deterministic linear response (`test_debye_huckel.py`, via the L-FTS prototype’s saddle): perturbing w_- by $\varepsilon \cos(k_0 x)$ reproduces $|\hat{c}| = \varepsilon \hat{h}^2 / (1 + E \hat{h}^2 / k_0^2)$ and $\hat{\psi} = E \hat{c} / k_0^2$ to 10^{-6} for $E \in \{0, 6.25, 25\}$ over six k_0 — pins the E normalization, $\hat{h}^2$ pair kernel and W_i assembly without any sampling. - Stochastic S_{cc} : with the exact single-mode Gaussian model (orig variables: $H_k = Bw^2 + \kappa\psi_o^2 - u_2(w + i\psi_o)^2$, $B = 1/\chi N$, $u_2 = \hat{h}^2/2$, $\kappa = k^2/2E$), the parameter-free predictions match: L-FTS $r(k) = \kappa^2(B - u_2)/[(\kappa + u_2)(B\kappa + Bu_2 - u_2\kappa - 2u_2^2)]$ agrees with 20k-step runs to 1–3% except the lowest- k shell; CL-FTS with the ANALYTIC estimator $\langle c_k c_{-k} \rangle$ (NOT $\langle |c_k|^2 \rangle$, which is not an observable in CL) follows $r(k) = (\kappa - B)(B - u_2)/[(B - u_2)(\kappa + u_2) + u_2^2]$ including its characteristic sign flip at $k = \sqrt{2EB}$; the strongest-signal shells agree to a few % ($E = 25$, $k_{\min}$: -3.08 vs -3.07). Residual deviations were pinned down quantitatively: (i) a finite- dt integrator bias — a synthetic single-mode replica of the exact discrete scheme reproduces the $dt=0.2$ run values and converges to the continuum prediction as $dt \rightarrow 0$, and real $dt=0.05$ runs move onto the synthetic curve (e.g. $k=4.71$: run -2.164 vs synthetic -2.172 , continuum -2.31); (ii) shells where the Gaussian signal crosses zero are dominated by a nonlinear (beyond-Gaussian) background that does not shrink with statistics — they are not usable for this comparison. - **One-loop DH thermodynamics PASSED (the CL-only validation):** the analytic observable $\langle \partial H / \partial E \rangle = -\langle H_{\text{exp}} \rangle / E$ measured over $E \in \{2.5, 6.25, 12.5, 25, 50\}$ (two Δt , linearly extrapolated) matches the exact-lattice Gaussian prediction $-(1/2\sqrt{n}V) \sum_{k \neq 0} (\kappa/E)(B - u_2) / \det(k)$ — the smeared generalization of the Debye–Hückel limiting law $-\kappa_D^3/12\pi$ — to $\leq 0.1\%$ at every coupling (ratios 0.999–1.001). This free energy comes entirely from ψ FLUCTUATIONS and is identically zero in the partial-saddle L-FTS. (Script: `dh_salt_runs/test_oneloop_dh.py`.) - Pitfalls recorded the hard way: $\chi N = 2$ IS the mean-field spinodal of the symmetric 1-segment salt (RPA breaks); cosine amplitudes on the cell-centered grid carry an $e^{ik dx/2}$ phase — compare $|\hat{c}|$; the exchange field’s explicit mass $1/\chi N$ caps the CL time step at $\Delta t < \chi N$; semi-implicit ψ integration biases per-mode variances (use ETD, above).

References

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5. *Macromolecules* **2025**, 58, 816 — the neutral multi-monomer framework (this library).